

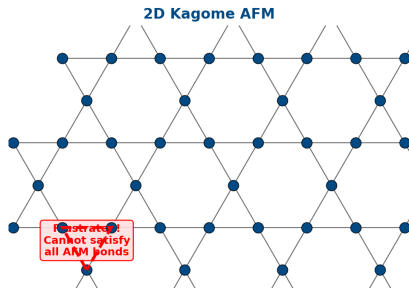
# Function-Free Optimization via Comparison Oracles

A Hypothetical Strategy for Frustrated Spin Systems

Learning Notes on arXiv 2604.26867v2

May 31, 2026

# Frustrated Spin Systems: Geometric Frustration



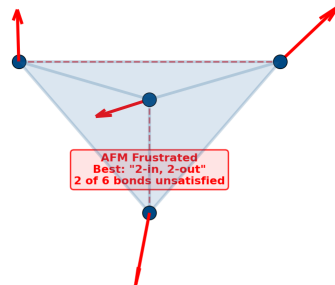
## 2D Kagome AFM

$$H = J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad J > 0$$

Cannot satisfy all 3 AFM bonds on a triangle

Geometric frustration  $\Rightarrow$  exponentially many  
near-degenerate minima  $\Rightarrow$  **rugged energy landscape**

## Pyrochlore AFM Tetrahedron



## 3D Pyrochlore AFM

Corner-sharing tetrahedra

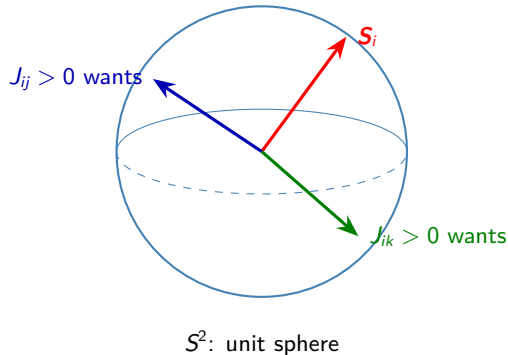
Best: "2-in, 2-out"  $\Rightarrow$  2/6 bonds unsatisfied

# Heisenberg Model: Continuous $S^2$ Configuration Space

- Heisenberg spin:  $\mathbf{S}_i \in S^2$  (unit sphere)
- Configuration space:  $(S^2)^N$
- Parameterize via spherical coordinates:

$$\mathbf{S}_i = (\sin \theta_i \cos \varphi_i, \sin \theta_i \sin \varphi_i, \cos \theta_i)$$

- Maps  $(S^2)^N \rightarrow \mathbb{R}^{2N}$  via  $(\theta_i, \varphi_i)$
- **Frustration**: competing interactions demand mutually contradictory spin orientations
- Coordinate singularities at  $\theta = 0, \pi$

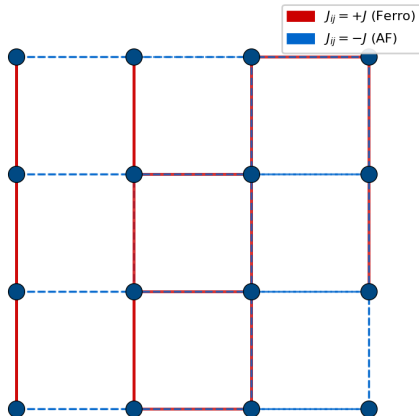


## Connection to FFO-CO

Continuous parameter space  $\mathbb{R}^{2N}$  is a prerequisite for FFO-CO.  
But frustrated systems violate *convex preference*  $\Rightarrow$  non-trivial extension required.

# Edwards–Anderson Model: Quenched Random Couplings

## 2D EA Model (Random $\pm J$ )



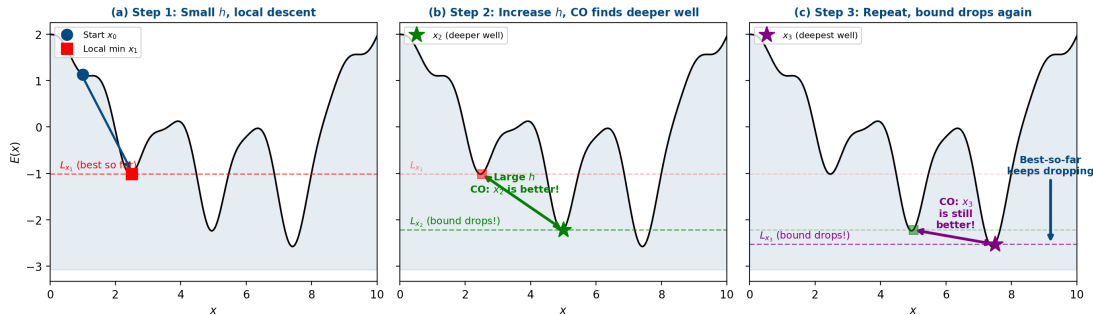
- Hamiltonian:

$$H = - \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$$

- $J_{ij}$  quenched random:  $\pm J$  (bimodal) or  $\mathcal{N}(0, \sigma^2)$  (Gaussian)
- **Rugged landscape:**  $\sim e^{cN}$  local minima
- Finding ground state: **NP-hard**
- Prototype of *spin glass* physics

# A Hypothetical FFO-CO Strategy for Rugged Landscapes

## FFO-CO Multi-Scale Strategy: Progressively Deeper Minima

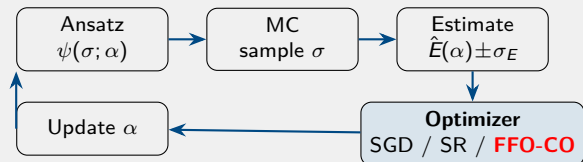


- 1 **Small  $h$ :** FFO-CO local descent  $\rightarrow$  local minimum  $x_1$
- 2 **Increase  $h$ :** comparison oracle discovers a *better* (deeper) region  $\Rightarrow$  level set  $L_{x_2}$  drops below  $L_{x_1}$
- 3 **Small  $h$  in new region  $\rightarrow$  local min  $x_2$ ; repeat  $\Rightarrow$  best-so-far bound keeps dropping**

**Key insight:** Comparison inconsistency (same direction, different  $h \Rightarrow$  flipped result) exposes *separated level sets* — structural information that gradients cannot provide.

# VMC: Application Scenario for FFO-CO

## VMC Pipeline



- Variational:  $E(\alpha) = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0$
- Jastrow:  $\psi = \exp(\sum_{i < j} \alpha_{ij} \sigma_i \sigma_j)$
- MC noise:  $\sigma_E \propto 1/\sqrt{N_s}$

## Noise: Gradient vs Sign

|        | SGD/SR                                       | FFO-CO                                               |
|--------|----------------------------------------------|------------------------------------------------------|
| Needs  | $\nabla E$                                   | $\text{sign}(\Delta E)$                              |
| Noise  | $\frac{\sigma_E}{h} \uparrow$<br>(amplified) | $P_{\text{ok}} > 97\%$<br>( $ \Delta E  > 3\sigma$ ) |
| Signal | $ \partial E / \partial \alpha $<br>(fixed)  | $ \Delta E $<br>(tunable via $h$ )                   |

**Key:** sign signal can be amplified by increasing comparison radius  $h$ ;  
gradient signal is fixed and its estimation requires dividing by  $h$ .

FFO-CO replaces only the **optimizer** — ansatz, sampler,  $E_{\text{loc}}$  computation are fully reused.

# Key Observations & Open Questions

## Comparison Oracle vs Gradient

- CO gives **global level-set ranking** between distant points
- Gradient is *purely local* — no relationship between distant gradients
- In noisy settings:  $\text{sign}(\Delta E)$  more robust than  $|\nabla E|$

## Comparison Consistency = Non-convexity Detector

- Same direction, different  $h$ :  
flipped comparison  $\Rightarrow$  non-convexity
- Binary search oscillation  
 $\Rightarrow$  level-set self-intersection
- Multi-direction contradictions  
 $\Rightarrow$  multi-basin structure

## Core Argument

“More information  $\neq$  better performance” — when gradient magnitude is unreliable (VMC, DMFT), using only *ordinal* information with adaptive  $h$  may outperform Adam-like GD variants.

Ref: K. Scheinberg & Z. Xiong, arXiv:2604.26867v2 (2026).